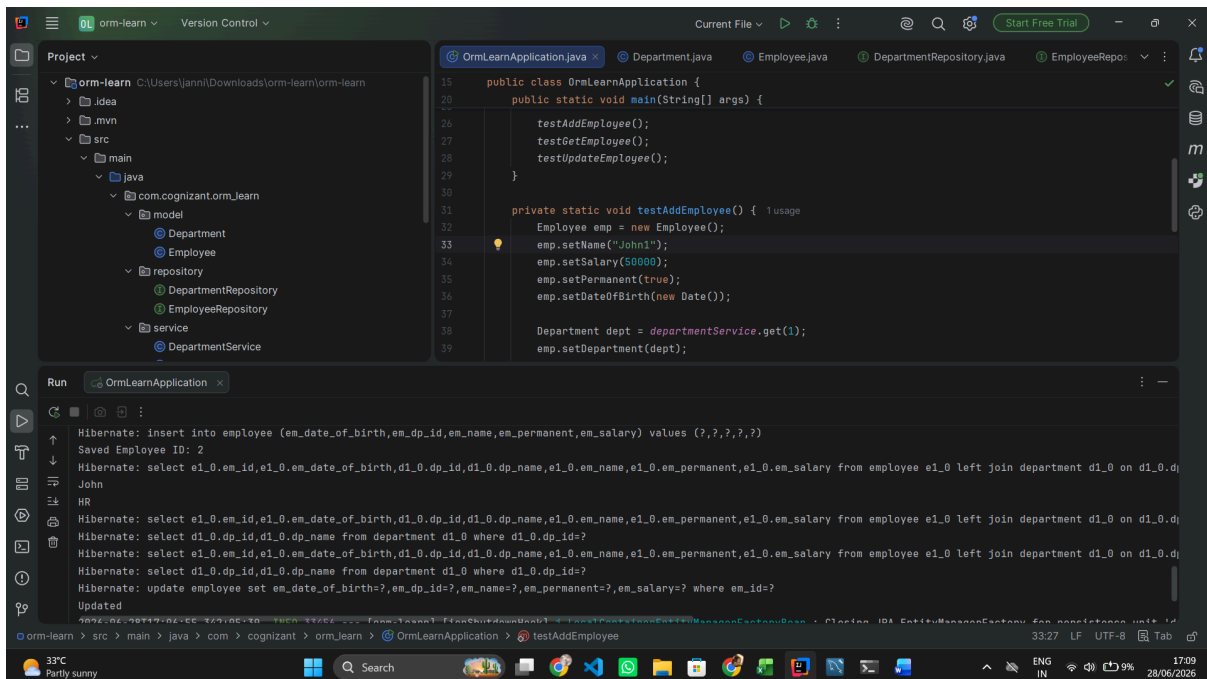


Output:

output:



The screenshot shows an IDE with the following components:

- Project Explorer:** Shows the project structure for 'orm-learn' with packages like 'com.cognizant.orm_learn', 'model', 'repository', and 'service'.
- Editor:** Displays the 'OrmLearnApplication.java' file. The code includes a 'main' method and a 'testAddEmployee' method. The 'testAddEmployee' method sets an employee's name to 'John1', salary to 50000, and permanent status to true, then saves it.
- Run Console:** Shows the output of the application. It includes Hibernate SQL statements for inserting and selecting employee data, and a message 'Saved Employee ID: 2'.

src/main/resources/application.properties

```
spring.application.name=orm-learn

spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
spring.datasource.username=root
spring.datasource.password=997662

spring.jpa.hibernate.ddl-auto=validate
spring.jpa.show-sql=true
```

src/main/java/com/cognizant/orm_learn/model/Employee.java

```
package com.cognizant.orm_learn.model;

import jakarta.persistence.*;
import java.util.Date;

@Entity
@Table(name = "employee")
public class Employee {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Column(name = "em_id")
    private int id;
```

```

@Column(name = "em_name")
private String name;

@Column(name = "em_salary")
private double salary;

@Column(name = "em_permanent")
private boolean permanent;

@Temporal(TemporalType.DATE)
@Column(name = "em_date_of_birth")
private Date dateOfBirth;

@ManyToOne
@JoinColumn(name = "em_dp_id")
private Department department;

public int getId() { return id; }
public void setId(int id) { this.id = id; }

public String getName() { return name; }
public void setName(String name) { this.name = name; }

public double getSalary() { return salary; }
public void setSalary(double salary) { this.salary = salary; }

public boolean isPermanent() { return permanent; }
public void setPermanent(boolean permanent) { this.permanent = permanent; }
}

public Date getDateOfBirth() { return dateOfBirth; }
public void setDateOfBirth(Date dateOfBirth) { this.dateOfBirth =
dateOfBirth; }

public Department getDepartment() { return department; }
public void setDepartment(Department department) { this.department =
department; }
}

```

src/main/java/com/cognizant/orm_learn/model/Department.java

```

package com.cognizant.orm_learn.model;

import jakarta.persistence.*;

@Entity
@Table(name = "department")
public class Department {

    @Id
    @Column(name = "dp_id")
    private int id;

    @Column(name = "dp_name")

```

```

private String name;

public int getId() { return id; }
public void setId(int id) { this.id = id; }

public String getName() { return name; }
public void setName(String name) { this.name = name; }
}

```

src/main/java/com/cognizant/orm_learn/repository/EmployeeRepository.java

```

package com.cognizant.orm_learn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import com.cognizant.orm_learn.model.Employee;

public interface EmployeeRepository extends JpaRepository<Employee, Integer>
{
}

```

src/main/java/com/cognizant/orm_learn/repository/DepartmentRepository.java

```

package com.cognizant.orm_learn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import com.cognizant.orm_learn.model.Department;

public interface DepartmentRepository extends JpaRepository<Department, Integer> {
}

```

src/main/java/com/cognizant/orm_learn/service/EmployeeService.java

```

package com.cognizant.orm_learn.service;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;
import com.cognizant.orm_learn.model.Employee;
import com.cognizant.orm_learn.repository.EmployeeRepository;

@Service
public class EmployeeService {

    @Autowired
    private EmployeeRepository employeeRepository;

    @Transactional
    public Employee get(int id) {
        return employeeRepository.findById(id).get();
    }

    @Transactional
}

```

```

    public void save(Employee employee) {
        employeeRepository.save(employee);
    }
}

```

src/main/java/com/cognizant/orm_learn/service/DepartmentService.java

```

package com.cognizant.orm_learn.service;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;
import com.cognizant.orm_learn.model.Department;
import com.cognizant.orm_learn.repository.DepartmentRepository;

@Service
public class DepartmentService {

    @Autowired
    private DepartmentRepository departmentRepository;

    @Transactional
    public Department get(int id) {
        return departmentRepository.findById(id).get();
    }

    @Transactional
    public void save(Department department) {
        departmentRepository.save(department);
    }
}

```

src/main/java/com/cognizant/orm_learn/OrmLearnApplication.java

```

package com.cognizant.orm_learn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

import com.cognizant.orm_learn.model.Employee;
import com.cognizant.orm_learn.model.Department;
import com.cognizant.orm_learn.service.EmployeeService;
import com.cognizant.orm_learn.service.DepartmentService;

import java.util.Date;

@SpringBootApplication
public class OrmLearnApplication {

    private static EmployeeService employeeService;
    private static DepartmentService departmentService;

    public static void main(String[] args) {

```

```

        ApplicationContext context =
SpringApplication.run(OrmLearnApplication.class, args);

        employeeService = context.getBean(EmployeeService.class);
        departmentService = context.getBean(DepartmentService.class);

        testAddEmployee();
        testGetEmployee();
        testUpdateEmployee();
    }

    private static void testAddEmployee() {
        Employee emp = new Employee();
        emp.setName("John1");
        emp.setSalary(50000);
        emp.setPermanent(true);
        emp.setDateOfBirth(new Date());

        Department dept = departmentService.get(1);
        emp.setDepartment(dept);

        employeeService.save(emp);

        System.out.println("Saved Employee ID: " + emp.getId());
    }

    private static void testGetEmployee() {
        Employee emp = employeeService.get(1);
        System.out.println(emp.getName());
        System.out.println(emp.getDepartment().getName());
    }

    private static void testUpdateEmployee() {
        Employee emp = employeeService.get(1);

        Department dept = departmentService.get(2);
        emp.setDepartment(dept);

        employeeService.save(emp);

        System.out.println("Updated");
    }
}

```